27 · The eye, part by part

The words of this track, as it says them. Bold brown is the examiner.

The eye, part by part. Her worksheets and her matching tables.

Her eye labelling and matching questions carry a lot of marks: one matching table alone is twenty marks, and her worksheets add about fifty more. The three layers track gives you the structure. This one walks through the eye from front to back, the way her worksheet describes each part, so you can name any part from its description.

The hook, front to back: conjunctiva, cornea, aqueous, iris and pupil, lens, vitreous, retina, choroid, sclera, and out through the blind spot.

The front. The conjunctiva is the thin mucous membrane that covers the front of the eye and lines the inside of the eyelids. It's a protective layer of epithelial cells, guarding the cornea against friction, and tears from the lacrimal glands keep it moist. It's what's infected in pinkeye. The cornea is the clear, curved window at the front, continuous with the sclera. Light is first bent here, and it does about eighty per cent of the refraction. Her worksheet adds that a cornea can be transplanted.

Behind the cornea is the anterior chamber, holding the aqueous humour, a clear, watery fluid that keeps the front of the eye in shape. It's drained by the canal of Schlemm, a circular canal where the sclera meets the cornea.

The iris is the coloured part: a pigmented muscular ring, with an inner ring of circular muscle, and an outer layer of radial muscle. It controls how much light gets in: enough to form an image, but not so much that it damages the retina. Her key on eye colour: people with blue eyes make lesser amounts of brown pigment in the iris. The pupil is the hole in the middle of the iris. It constricts in bright light, and dilates in dim light, under autonomic control.

The middle. The ciliary body holds the lens in place with its suspensory ligaments, secretes the aqueous humour, and contains the ciliary muscles, which are involuntary, and change the shape of the lens for near and far. That's accommodation. The lens itself is transparent, flexible and biconvex, and does the other twenty per cent of the refraction.

Behind the lens is the vitreous humour: a transparent, jelly-like, protein-rich mass. It maintains the shape of the eyeball, the back part, and her worksheet says it provides nutrients. A small, transparent canal runs through it, from the optic disc to the lens: the hyaloid canal.

The back. The retina is the sensory layer, and her key says the sensory layer is the one responsible for vision. Its rods and cones turn light into electrical impulses, and the image forms here. Cones for colour vision, rods for poor light. At its centre is the macula lutea, an oval yellow spot. In the middle of that is the fovea centralis, a small pit directly in line with the pupil, with only cones: sharp colour vision in good light.

The optic disc, the blind spot, is the exit point for the ganglion cell axons, where they leave as the optic nerve. It has no rods or cones, so no image forms there. The optic nerve carries the impulses to the occipital lobe, which controls vision.

The choroid is the black layer between the retina and the sclera. Its dark pigment stops light reflecting around inside the eyeball, and its network of blood vessels feeds the retina and removes waste. And the sclera is the tough, white, non-elastic outer coat. It protects the eye, keeps it round with the pressure of the fluid inside, and anchors the muscles that move it.

Her one-mark eye facts. Of all the senses, vision has the majority of the body's sensory receptors. Strabismus is when a person can't align both eyes at the same time. Retinal detachment is when the retina tears and comes away, and it's treated with laser. Myopia, in her tables: the lens is too strong, or too bulgy, or the eye is too long, so the light is over-focused, in front of the retina, and a biconcave lens scatters it back. Hyperopia: the lens is too weak, or too flat, or the eye is too short, so the light focuses behind the retina, and a biconvex lens boosts it.

Two words from her short questions. An afterimage is an image that persists after you close your eyes. Her key says it's thought to be due to a vivid stimulus blocking the photoreceptors, so they can't fire again straight away. And an inverted image has two meanings in her quizzes. In her lens drop-down, inverted means upside down: the biconvex lens makes a real, inverted image, and the brain turns it upright. In her short-answer quiz, her model answer means an image in opposite colours: stare at it, look away, and the afterimage shows the true colours.

The traps. The aqueous humour is at the front, the vitreous humour is behind the lens. The optic disc is the blind spot; the fovea is the sharpest spot. And the choroid is black on purpose: it stops light bouncing around inside the eye.

Once more. The eye, part by part.

The conjunctiva covers the front of the eye and lines the eyelids, and it's infected in pinkeye. The cornea is the clear window where light is first bent: eighty per cent of the refraction, and it can be transplanted. The aqueous humour fills the anterior chamber and drains through the canal of Schlemm. The iris has circular and radial muscle, and blue eyes have less brown pigment. The ciliary body holds the lens, makes aqueous humour, and its involuntary muscles change the lens shape. The lens is biconvex: twenty per cent of the refraction. The vitreous humour is the jelly behind the lens, with the hyaloid canal through it.

The retina is the sensory layer, where the image forms. The macula lutea is the yellow spot, and the fovea, in line with the pupil, has only cones. The optic disc, the blind spot, is where the axons leave as the optic nerve, to the occipital lobe. The choroid is black, to stop reflection, and feeds the retina. The sclera is tough and white, and keeps the eye round. Vision has most of the body's receptors. Strabismus: the eyes can't align. Myopia: too bulgy a lens, biconcave fix. Hyperopia: too flat, biconvex fix.